

Silent voice: AI Based Wearable Glove for Sign Language Translation



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In the name of Allah (SWT), the most beneficent and the most merciful

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Computer Engineering International Islamic University, Islamabad
In partial fulfillment of the requirements for the award of the degree of
Bachelor of Science in Computer Engineering Technology

Declaration

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Acknowledgments

This BS thesis in Computer Engineering Technology has been conducted at Department of Electrical and Computer Engineering, Faculty of Engineering and Technology, International Islamic University, as part of the degree program. We would like to thank Dr. Ubaid Umar for providing us an opportunity to work on this project, under his supervision and guidance throughout the project. We would also like to thank Dr. Maria Murad for her help, efforts and dedicated support throughout the project. We are also particularly thankful to Almighty Allah and grateful to our parents, brothers, and sisters who always supported and encouraged us during our project and studies at IIUI.

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Date Started: September, 2025

Date Completed: May, 2026

Tools Used:

- Android Studio
- Google Colab
- Arduino

Abstract

We developed an American Sign Language (ASL) gesture to text/speech real-time wearable assistive communication system (Silent Voice). The scope of the work included the design of a sensor glove instrumented with 5 number of resistive flex sensors and one MPU6050 Inertial Measurement Unit (IMU) and powered by an Arduino Nano microcontroller; then, the hand gesture data, comprising of 11 features, was wirelessly sent to an Android smartphone paired with the sensor glove through HC-05 Bluetooth Classic module. The sensor data is sent to a dedicated Android application that is implemented in Kotlin, 100 frames are buffered in each window and an on-device inference is made using a two-layer Long Short-Term Memory (LSTM) neural network that was compiled in the TensorFlow Lite format to classify 30 ASL Gesture classes (A–Z and common words: yes, no, me, you). On-screen text appears for gestures that are recognized, which are vocalized with the built-in Android Text-to-Speech engine and which are written in a local Room database to link them to the time of occurrence. The model LSTM was trained on the publicly shared ASL Sensor Dataglove Dataset on the platform google colab, which gave a test accuracy of 77.93% and a macro average of the F1 score of 0.76 for all the classes. The system does not need an external camera, internet connection and server to process the data, and can be used as a mobile and economical assistive tool for deaf and hard-of-hearing people almost anywhere.

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List of Abbreviations

SoC	System on Chip
FYP	Final Year Project
OBE	Outcome Based Education
IoT	Internet of Things
ASL	American Sign Language
IMU	Inertial Measurement Unit
LSTM	Long Short-Term Memory
TFLite	TensorFlow Lite
BLE	Bluetooth Low Energy
TTS	Text-to-Speech
SLR	Sign Language Recognition
RFCOMM	Radio Frequency Communication

Chapter 1

Introduction

This chapter aims to provide motivation for the development of the Silent Voice Sign Language Translation Glove, offer an overview of the project, lay out the problem statement, the project objectives, the methodology considered, and an outline of this report.

1.1 Motivation

Communication is a basic human right, but for the estimated 430 million individuals who suffer a disabling loss of hearing all over the world, effective communication with the hearing population (community) is still a continuous problem [1]. A huge communication gap exists between the hearing and the deaf and hearing impaired community with the majority of the hearing community not knowing sign language while it is the primary mode of expression for the deaf and hard of hearing community in daily life, healthcare, education and the workplace.

The current assistive technologies are either vision based and require controlled light systems and specific cameras, or they are only able to detect a small number of signs. Definitely there's a need for a portable, low cost, real time solution to be worn normally, and used anywhere. To fill this gap, the Silent Voice project aims to create a sensor based wearable glove which is connected to android mobile application that converts American Sign Language (ASL) gestures to the text and speech in real-time without the need of camera or internet connection.

1.2 Project Overview

Silent Voice is an assistive technology system to convert (ASL) hand movements in realtime into text and speech using a wearable device. The system is comprised of two components that are closely integrated together, namely, a smart sensor glove and an Android App, the Silent Voice. The glove has six Resistive Flex Sensors (one for each finger) to measure the bending and an MPU6050 Inertial Measurement Unit (IMU) six axis motion sensor detects the six axes of motion data (three axis gyroscope and three axis accelerometer). All sensor values are read by the Arduino Nano microcontroller and the values are sent wirelessly to an Android smart phone paired with the HC-05 Bluetooth Classic to which the microcontroller is connected.

The sensor data is then passed to the Silent Voice Android app which buffers 100 frames of the sensor data and passes it to an on-device model implemented using a TensorFlow Lite Long Short-Term Memory (LSTM) neural network, which classifies the ASL being performed. The gestures that are recognized will appear on the screen and will be translated into english and voiced by Android's on-board Text to Speech engine. All recognizable gestures are saved into a local Room database with timestamps and user preferences like speech rate and pitch can be set via a settings screen.

1.2.1 Key Features of the System

The key features of the system are:

- ASL real-time recognition for 30 classes (A–Z plus yes, no, me, you)
- 5 Flex sensors and MPU6050 IMU sensor on the hand (worn by the individual)
- A microcontroller, Arduino Nano for data acquisition on the glove
- Wireless communication using Bluetooth classic (HC-05) between Glove and smart phone
- Inference using an on-device TensorFlow Lite model (no connection to internet required).
- Speech, Braille, and/or enlarged print feedback in English

All of the above plus gesture history log with timestamps stored in Room database

- Speakable rate, pitch, history management and font size can be set

1.3 Problem Statement

Today there is still no easy bridge over the communication gap between the Deaf and Hard of Hearing community and the mainstream hearing community through current mainstream communications technology. Vision based sign language recognition systems are sensitive to the lighting condition and background noises, and are not applicable to everyday portable uses. Currently, the systems available using a glove either are very expensive research systems or have a very small number of signs (<10) or rely on server-side processing and a network connection. Thus, an inexpensive, practical and wearable system that is able to detect full vocabulary ASL in real time, without the need of a camera, internet connection or even costly hardware parts is in demand.

1.4 Project Objectives

This project is aimed at the following objectives:

- To build and design the wearable glove with five flex sensors and an MPU6050 IMU to collect the flex and orientation data for all the fingers for identifying ASL gestures.
- To accomplish a secure real-time wireless communication between the glove and a Android cellphone with the HC-05 Bluetooth Classic module.
- To construct and train a deep learning model based on LSTM with dual layers to recognize 30 ASL gesture classes (26 letters A–Z and 4 common words: yes, no, me, you) using the time-series sensor data with acceptable accuracy.
- To deploy the trained model as a TensorFlow Lite file on an Android device and efficiently perform on-device inferences without Internet access.
- To create an Android application named Silent Voice that can process the data from sensor and execute LSTM model and display the recognized gestures in the users' phones as a text and also speech in English.

To validate the system end-to-end with testing on inputs of Bluetooth communication, Model Inference, Gesture Recognition accuracy and reliability of system.

1.5 Brief Project Methodology

The project was delivered as 4 workstreams in parallel. The hardware team designed and assembled the sensor glove, they then selected and calibrated the flex sensors and MPU6050 IMU selected, and programmed the Arduino Nano to sample and send the 11 features sensor data frames via Bluetooth. The software team created the Silent Voice Android app using Kotlin, introducing the Bluetooth RFCOMM communication stack, data parsing & buffering pipeline, and the TensorFlow Lite inference engine. Concurrently, the machine learning team gathered and preprocessed the ASL Sensor Dataglove Dataset, developed and trained the model LSTM on Google Colab, tested the model and finally exported it to the TensorFlow Lite format. Finally, the overall system was integrated together in a single place and an end-to-end test was performed with the physical gloves and the Android application with a number of test cases covering Bluetooth connectivity, packet parsing, model inference, confidence filtering, and TTS output.

1.6 Report Outline

The rest of this report is structured as follows. A literature survey of sign language recognition system and both vision based and glove based recognition system are done in chapter-2 and in chapter-2 contribution of this project has been identified. Chapter 3 outlines the detailed design and implementation of the entire system, from hardware configuration to software architecture, dataset pre-processing, LSTM model training, and the Bluetooth communication protocol, to the real-time prediction pipeline. Testing and validation are described in Chapter 4, and the results are presented by test cases as well as the accuracy of the model as iterations increased and per-class classification results. Finally, Chapter 5 provides an overview of the successful outcomes and suggestions for further research.

Chapter 2

Literature Review

This chapter reviews existing research and systems related to sign language recognition, with a focus on wearable glove-based approaches and deep learning techniques relevant to this project. Section 2.1 provides background on the problem domain and the two main categories of sign language recognition systems. Section 2.2 reviews related work and recent publications. Section 2.3 identifies the specific contribution of the Silent Voice project. Section 2.4 summarizes the chapter.

2.1 Background of Project

Sign language recognition (SLR) systems can be broadly categorized into two approaches: vision-based systems and sensor-based (glove-based) systems. Vision-based systems use cameras and computer vision techniques to detect and classify hand gestures from video frames. Sensor-based systems use wearable instrumented gloves to directly measure finger bending, hand orientation, and motion.

Vision-based approaches have gained popularity due to the widespread availability of cameras and frameworks such as MediaPipe. Using MediaPipe's hand-tracking capabilities, researchers have demonstrated accuracy as high as 99.12% on ASL datasets using vision-based methods. However, vision-based systems are highly sensitive to lighting conditions, background clutter, and camera angle, and are impractical for everyday portable use outside controlled environments.

Sensor-based glove systems overcome these limitations by capturing hand kinematics directly. Flex sensors measure the bending angles of individual fingers, while Inertial Measurement Units (IMUs) capture the orientation and motion of the hand. These measurements are stable regardless of lighting or background, making glove-based systems more robust for real-world deployment. The challenge with glove-based systems lies in designing accurate classifiers for the time-series sensor data they produce, which is where deep learning — particularly recurrent architectures — has shown strong results.

2.2 Related Work

2.2.1 Glove-Based ASL Recognition with Deep Learning

In their work, Dalal et al. (2024) proposed a smart glove with a similar hardware setup consisting of five flex sensors and an MPU6050 IMU for real-time ASL translation using deep learning techniques [3]. They used their system with a Recurrent Neural Network trained with their own collected dataset of 27 classes of ASL gestures with real time inference. The study Recognizing sensitivity to small vibrations as a significant problem with the MPU6050, bit-shifting methods were

suggested for stabilizing the readings of the IMS. A similar system was introduced by Amin et al. (2025) who created a smart hand glove for ASL recognition using flex sensors and a 9-DOF IMU, and exploited a bidirectional LSTM model [4]. Their system, which is also based on an Arduino Nano microcontroller, was trained with a dataset of 26 classes of ASL alphabet and showed the suitability of LSTM based system to capture the temporal dynamics of sign language gestures from wearable sensor data. Maharjan et al., (2024) proposed a smart glove for real-time sign language detection and translation using flex sensors and an LSTM based RNN [5]. They were based around the ASL letters A to E and the numbers 0 to 9 and also had a Text-to-Speech engine to generate both the text on the screen and the voicespeak output, thus following the same design principle as the Silent Voice system.

2.2.2 Sequential Gesture Data using LSTM

At the IC3-2024 conference, Bhadouria et al. (2024) introduced an LSTM-based sign language recognition system, which shows that LSTM networks can be used efficiently with MediaPipe for gesture classification [6]. They demonstrated that their LSTM models could learn the long-term temporal dependency of the hand motion sequences, which is crucial for hand motion time-series data that has limited temporal dependency, unlike more complex data that can be classified by Support Vector Machines or feedforward neural networks. Kalandar and Dworakowski (2023) presented a wearable system for interpreting a subset of dynamic words in ASL based on the data recorded by three flex sensors as time sequences and machine learning [7]. Their proof-of-concept showed that even a limited number of flex sensors, along with proper sequential modelling, can provide useful discrimination of gestures, which is why they decided to use 5 flex sensors in conjunction with the IMU in this project.

2.2.3 IoT-Enabled Glove with Arduino Nano

The study recently published in International Journal of Innovations in Science and Technology (2025) proposed an IoT based smart glove which uses five flex sensors and an MPU-6050 gyroscope, and the processing is done using an Arduino Nano microcontroller [8]. This system is the closest to the Silent Voice hardware configuration. The study showed that the gesture classification was achieved successfully in real-time and that the Arduino Nano was appropriate for realization of a low power, small glove — thus justifying the hardware decisions in this project.

2.2.4 Wearable Inertial Sensor-Based ASL Recognition

Ji et al. (2023) designed a sign language recognition system based on the inertial sensors of a dataglove worn on a person, and achieved excellent results in ASL gesture classification with data collected from the hearing and speech impaired [9]. They highlighted that it was crucial to incorporate a diverse set of users in the training data to ensure generalization. Abdullahi and Chamnongthai (2022) also showed the word recognition of ASL through multi-stacked deep LSTM networks on skeletal video data and obtained similar results, which corroborated the suitability of stacked LSTM networks in the sign language classification task [10].

2.2.5 Deep Learning Approaches for Sign Language Recognition

The use of deep learning techniques has been widely used in recognition system for sign language which has been significantly enhanced its performance in automatic learning of complex patterns from sensor and visual information. In traditional machine learning approaches, like Support Vector Machines (SVM) and Random Forest, features must be extracted manually, whereas deep learning models can learn meaningful representations directly from the raw sequence(s) of data. Due to the capability of extracting spatial features from hand images, Convolutional Neural Networks (CNNs) have been widely used for sign language recognition based on images. But, CNN-based methods typically demand large volumes of images and regulated camera settings. In the case of wearables glove-based systems, sequential models like Recurrent Neural Networks (RNNs) and LSTM networks are more appropriate, since the gestures in sign language are continuous, implying that the palm of the hand is changing over time and the fingers are moving continuously. To overcome the drawbacks of traditional RNNs, LSTM networks are introduced, which can remember the previous time steps and thus overcome the problem of long-term dependence of gesture sequences. This renders LSTM-based models efficient in data processing of flex sensors and IMUs. The similar approach has been taken in the Silent Voice project where the ASL alphabet gestures are recognized from real-time sensor signals from the glove using LSTM-based classification.

2.2.6 Real-Time Mobile-Based Sign Language Translation Systems

Recent studies have centered on the design of portable and easy to use sign language translation systems which can be used in the real world. The benefits of mobile-based solutions include being portable, cost reduction and being independent of external computers. Using frameworks like TensorFlow Lite, machine learning models can be deployed on mobile devices, enabling real-time prediction without the need for cloud connectivity. There have been several studies that have investigated the combination of wearable sensors and mobile applications for assistive communication. These systems include sensor information acquisition, gesture classification and speech generation in a single system. Many of the current solutions, however, have a limited vocabulary of gestures, require Internet services, or don't have a complete user interface. The Silent Voice system consists of a smart glove developed on the basis of the Arduino Nano microcontroller

and an Android application which can recognize the ASL alphabet in real-time, store the gestures history and convert the text to speech. Full on-device processing guarantees privacy, reliability and usability in everyday communications.

2.2.7 Sign Language Recognition using Wearable Sensor Fusion

Wearable sign language recognition systems typically utilize a combination of various sensors to enhance the accuracy and reliability of sign language recognition. Flex sensors are used to gather information about the bending angles of each individual finger and IMUs (Inertial Measurement Units) are used to capture hand orientation, rotation, and movement patterns. These sensors collectively provide a better representation of hand gestures than using any one sensor type alone. A lot of human activity and gesture recognition systems applied sensor fusion techniques in order to overcome the drawbacks associated to individual sensors. For instance, the flex sensors can accurately report the position of each finger, but not the movement of the hand, while the IMU can accurately report the movement of the hand, but not the position of the fingers. Wearable glove systems can perform better in the recognition of complex gestures by using both sources of information. The Silent Voice project follows a similar sensor fusion concept where it uses five flex sensors in combination with the MPU6050 IMU. This helps the system to robustly recognize ASL alphabet in real time when the fingers' positions are static and the hand moves dynamically.

2.2.8 Existing Glove-Based Systems Challenges and Limitations

Significant progress has been made in the development of wearable sign language recognition devices using gloves but various issues have to be addressed. There is an obvious drawback that the data sets need to be large and varied, as sensor readings can vary from one user to another, depending on the size of the hand, the style of gesture and the speed of the movement. Systems developed with a small amount of data can be accurate within the controlled setting, but are less accurate when used in a real setting. A further issue is to balance accuracy with processing speed and cost of hardware. Expensive sensors or external computing systems are required for high performance systems, which limits their practicality for everyday use as an assistive system. Furthermore, real-time systems need to overcome noise from the sensors, unintended hand motions, and predictions that are wrong. Silent Voice tackles these problems with a low cost hardware design, real-time filtering techniques on the sensor, confidence based prediction, and on-device TensorFlow Lite inference. Such enhancements contribute to the development of an accessible and convenient communication service for people who need sign language support.

2.2.9 Collecting Data and Real-Time Gesture Recognition in Wearable Systems

The quality and diversity of the input sign language dataset used for training are significant for the performance of sign language recognition systems. Wearable systems with sensors mounted on gloves capture a time-series of sensor data including the bending of the fingers and hand motion data. In contrast to image-based datasets, datasets generated by the sensors capture the continuous

evolution of hand posture and hand movements, and thus require suitable preprocessing and sequential learning approaches. Some research works have emphasized the need to gather gesture samples from various users to improve the generalization of the model, and from different execution speeds to account for variations in speed. When the system is used by new users, it may be overfitted, and recognition accuracy may be low due to a limited dataset. Hence, in order to increase prediction stability in real-time sign language systems, data normalization, data filtering, and learning based on sequences of data are widely used. The Silent Voice project does just this – it gathers sequences from flex sensors and MPU6050 readings in real-time. Rolling input buffer and confidence based prediction mechanism enables keeping gesture recognition stable and decrease false classification in practical use.

2.2.10 Deployment of Machine Learning Models on Embedded and Mobile Platforms

The latest advancements in the field of assistive technology have been in the direction of on-device deployment of machine learning models for portable and efficient solutions. Cloud-based recognition systems can offer high computational power but come with their own set of challenges, such as latency, internet connectivity, and privacy concerns. On-device inference enables systems to function autonomously in real-world settings and offers quicker response time. The frameworks, like TensorFlow Lite, include the ability to efficiently execute trained deep learning models on smartphones and embedded devices that have limited computational capability. This method can be used in a wearable sign language recognition system where low power consumption, portability and real time response are essential. The Silent Voice system is based on this idea and has transformed the trained recognition model into TensorFlow Lite format and then wrapped it into an Android application. This enables real-time ASL alphabet prediction, gesture history management and Text-to-Speech output without the need of a server, or constant internet access.

2.3 Project Contribution

The Silent Voice project contributes the following over the existing work:

- A full end-to-end system from the low-cost Arduino Nano glove to the dedicated Android app — inference is all on-device and no server, cloud, or Internet connection is needed.
- 30 ASL classes (A–Z plus yes, no, me, you) recognition with a wider vocabulary than most of the similar low cost glove-based systems.
- Development of a strong real-time prediction pipeline on Android that includes a 100-frame rolling buffer, a flex sensor variance based resting-hand motion filter, a 75% confidence threshold, and a three-consecutive-confirmation rule to reject false predictions.
- Building of a complete Android app (Silent Voice) for gestures history recording, voice output via TTS and settings configuration interface, thus turning the system into something more than a research prototype that can be used in practice as an assistive technology system.

2.4 Summary

The present chapter presented a thorough literature survey on sign language recognition systems, which includes vision-based, glove-based, LSTM based sign language sequential classifiers and IoT based wearable sign language recognition systems. It is concluded from the review that the glove based system is well established and is effective with the flex sensors and IMUs using LSTM model for real time ASL recognition. The Silent Voice project extends from this by integrating a low cost Arduino Nano based glove with a custom made Android App and on device TensorFlow Lite inference, creating a functional, portable, camera-less assistive communication system. A full system design and implementation is described in the following chapter.

Chapter 3

System and Implementation Details

This chapter brings the entire system of the Sign Language Translation Glove project, its design and implementation details. The software is divided into 3 modules: (1) Silent Voice Android App, (2) ASL Gesture Recognition TensorFlow Lite Machine Learning Model, and (3) Bluetooth Communication Module that connects the glove hardware to the Android App.

3.1 System

The system can be divided into three stages: (1) Sensor Data Acquisition from the glove, (2) Wireless Data Transmission (Using Bluetooth Classic, RFCOMM), (3) On-Device Inference (Using TensorFlow Lite). The glove continually reads the values of the sensors, and transmits them as a string of comma separated packets. These are then parsed by the app and the 100 frames are stored in a rolling buffer, while the LSTM model is executed every 10 new frames, thus generating a real-time prediction.

3.1.1 System Architecture

The system can be divided into three stages: (1) Sensor Data Acquisition from the glove, (2) Wireless Data Transmission (Using Bluetooth Classic, RFCOMM), (3) On-Device Inference (Using TensorFlow Lite). The glove continually reads the values of the sensors, and transmits them as a string of comma separated packets. These are then parsed by the app and the 100 frames are stored in a rolling buffer, while the LSTM model is executed every 10 new frames, thus generating a real-time prediction.

3.1.2 Requirements Analysis

The software part of the system must meet the following criteria:

- The application must connect to the HC-05 Bluetooth module using SPP UUID (00001101-0000-1000-8000-00805F9B34FB) via RFCOMM socket.
- Android Bluetooth permissions must be handled: Android 11 and below: BLUETOOTH, BLUETOOTH_ADMIN and Android 12 and above: BLUETOOTH_CONNECT, BLUETOOTH_SCAN.
- Each frame will require 11 float values: 5 flex sensors (flex_1 – flex_5), 3 gyroscope (GYRx, GYRy, GYRz) and 3 accelerometer (ACCx, ACCy, ACCz).
- To provide temporal context for the LSTM model, at least 100 frames (samples) are buffered before trying to infer.
- The model should have a confidence score of at least 75% before it makes a prediction. To avoid false positives, the same gesture will have to be confirmed three times in a row before it is displayed and spoken.

- The application should include Text-to-Speech in the English language.
- The Room's history of gestures should be saved in a local Room database with timestages.
- All user preferences such as the TTS language, speech rate, pitch, text size and history preferences should be adjustable on a Settings screen. The sequence of the task is determined by the first launch. The order of the task is decided at the initial launch.

The following section describes how to set up the hardware and development environment.

The hardware prototype was built on a perfboard with the use of an Arduino Nano microcontroller.

The Arduino Nano was selected for being small enough to fit on the glove, having eight channels of analog input, built-in I2C on pins A4 (SDA) and A5 (SCL) and being compatible with the CH340 USB-to-serial chip for programming via USB. Arduino IDE 2.x and CH340 Windows driver were used for development.

The HC-05 Bluetooth module was programmed with AT commands first with baud rate 9600. D10, D11 were used as SoftwareSerial to provide a second UART with D0/D1 used for USB serial debugging. Power is delivered via a 3.7V 18650 Li-ion battery, a USB-C charging module (TP4056) and a DC-DC boost converter that regulates 5V to the Vin pin of the Arduino Nano and all 5V peripheral devices.

3.2 Methodological and Implementation Details

3.2.1 Hardware and Development Setup

The hardware and development setup is described below. Hardware prototype was built on perfboard with the main microcontroller being the Arduino Nano. The Arduino Nano was selected for its small footprint for mounting in the glove, 8 analog inputs, built-in I2C support on dedicated pins A4 (SDA) and A5 (SCL), and the ability to be programmed via USB through the CH340 USB-to-serial chip. The development environment consisted of Arduino IDE 2.x and the CH340 Windows driver was installed. The HC-05 Bluetooth module was programmed in AT command mode with a baud rate of 9600. D10 and D11 were used with SoftwareSerial to make a second UART to keep D0/D1 free for USB serial debugging. The power comes from a 3.7V 18650 Li-ion cell, managed by a TP4056 USB-C charging module and a DC-DC boost converter which supplies a regulated 5V output to the Arduino Nano VIN pin and all 5V peripherals.

3.2.2 Hardware Details

This section describes each hardware component, its operating principle, electrical characteristics, and specific role in the ASL Sensor Glove system.

i. Arduino Nano

The Arduino Nano is a compact microcontroller board based on the ATmega328P, running at 5V and 16MHz. It provides 8 analog inputs (A0–A7), 14 digital I/O pins, hardware I2C on A4/A5, and hardware UART on D0/D1. Five analog pins (A0–A3, A6) read the flex sensors. A4 and A5 are

dedicated to MPU6050 I2C. D10 and D11 are configured SoftwareSerial for HC-05. The CH340 USB bridge requires a Windows driver before the board is recognized as a COM port.



Figure 1 Arduino nano

Code

```
#include <Wire.h>
#include <SoftwareSerial.h>

// ----- BLUETOOTH -----
SoftwareSerial BT(11, 10);

// ----- MPU6050 -----
const int MPU_ADDR = 0x68;
int16_t accX, accY, accZ;
int16_t gyroX, gyroY, gyroZ;

// ----- CALIBRATION OFFSETS -----
float flex_rest[5] = {0};
float gyroOffset[3] = {0};
float accOffset[3] = {0};

// ----- SCALING FACTORS -----
float flex_scale[5] = {-1.58, -1.04, -1.34, -1.95, -1.39};

const float GYRO_SCALE = 1.0 / 131.0;
const float ACC_SCALE = 9.81 / 16384.0;

// ----- FLEX -----
int flexPins[5] = {A7, A6, A3, A2, A1};
float flexSmooth[5] = {0};
float alpha = 0.3;

// ----- TIMING -----
unsigned long lastSend = 0;
const int sendInterval = 50;

// =====
void readMPU() {
  Wire.beginTransaction(MPU_ADDR);
  Wire.write(0x3B);
  Wire.endTransmission(false);
```

```

Wire.requestFrom(MPU_ADDR, 14, true);

accX = Wire.read() << 8 | Wire.read();
accY = Wire.read() << 8 | Wire.read();
accZ = Wire.read() << 8 | Wire.read();
Wire.read(); Wire.read();
gyroX = Wire.read() << 8 | Wire.read();
gyroY = Wire.read() << 8 | Wire.read();
gyroZ = Wire.read() << 8 | Wire.read();
}

// =====
void calibrate() {
  // FIX 1: Clear instructions + 5 second delay
  Serial.println(">>> GET READY! Open hand STRAIGHT! <<<");
  Serial.println(">>> Palm facing WALL! Fingers STRAIGHT! <<<");
  BT.println("CALIBRATING");
  delay(5000); // 5 seconds to position hand

  Serial.println(">>> Calibrating NOW... Hold STILL! <<<");

  long gxSum=0, gySum=0, gzSum=0;
  long axSum=0, aySum=0, azSum=0;
  long flexSum[5] = {0};
  int samples = 100;

  for (int i = 0; i < samples; i++) {
    readMPU();
    gxSum += gyroX;
    gySum += gyroY;
    gzSum += gyroZ;
    axSum += accX;
    aySum += accY;
    azSum += accZ;

    for (int j = 0; j < 5; j++) {
      flexSum[j] += analogRead(flexPins[j]);
    }
    delay(10);
  }

  gyroOffset[0] = gxSum / samples;
  gyroOffset[1] = gySum / samples;
  gyroOffset[2] = gzSum / samples;
  accOffset[0] = axSum / samples;
  accOffset[1] = aySum / samples;
  accOffset[2] = azSum / samples;

  for (int j = 0; j < 5; j++) {
    flex_rest[j] = flexSum[j] / samples;
  }

  Serial.println(">>> Calibration Done! <<<");
  Serial.print("Flex Rest: ");
  for (int j = 0; j < 5; j++) {
    Serial.print(flex_rest[j]); Serial.print(" ");
  }
  Serial.println();
}

```

```

}

// =====
void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  Wire.begin();

  Wire.beginTransmission(MPU_ADDR);
  Wire.write(0x6B);
  Wire.write(0);
  Wire.endTransmission(true);

  delay(200);
  calibrate();

  Serial.println("System Ready");
  BT.println("START");
}

// =====
void loop() {
  if (millis() - lastSend < sendInterval) return;
  lastSend = millis();

  // ----- FLEX -----
  for (int i = 0; i < 5; i++) {
    int raw = analogRead(flexPins[i]);
    flexSmooth[i] = alpha * raw + (1 - alpha) * flexSmooth[i];
  }

  // ----- MPU -----
  readMPU();

  // ----- SCALE FLEX -----
  float calFlex[5];
  for (int i = 0; i < 5; i++) {
    calFlex[i] = (flexSmooth[i] - flex_rest[i]) * flex_scale[i];
    // FIX 2: Clamp values to stop crazy readings
    calFlex[i] = constrain(calFlex[i], -50, 400);
  }

  // ----- SCALE IMU -----
  float calGyrX = (gyroX - gyroOffset[0]) * GYRO_SCALE;
  float calGyrY = (gyroY - gyroOffset[1]) * GYRO_SCALE;
  float calGyrZ = (gyroZ - gyroOffset[2]) * GYRO_SCALE;

  float calAccX = (accX - accOffset[0]) * ACC_SCALE;
  float calAccY = (accY - accOffset[1]) * ACC_SCALE;
  float calAccZ = (accZ - accOffset[2]) * ACC_SCALE;

  // ----- SEND PACKET -----
  BT.print("<");
  BT.print(calFlex[0],2); BT.print(",");
  BT.print(calFlex[1],2); BT.print(",");

```

```

BT.print(calFlex[2],2); BT.print(",");
BT.print(calFlex[3],2); BT.print(",");
BT.print(calFlex[4],2); BT.print(",");
BT.print(calGyrX,4); BT.print(",");
BT.print(calGyrY,4); BT.print(",");
BT.print(calGyrZ,4); BT.print(",");
BT.print(calAccX,2); BT.print(",");
BT.print(calAccY,2); BT.print(",");
BT.print(calAccZ,2);
BT.println(">");

// ----- DEBUG -----
Serial.print("<");
Serial.print(calFlex[0],2); Serial.print(",");
Serial.print(calFlex[1],2); Serial.print(",");
Serial.print(calFlex[2],2); Serial.print(",");
Serial.print(calFlex[3],2); Serial.print(",");
Serial.print(calFlex[4],2); Serial.print(",");
Serial.print(calGyrX,4); Serial.print(",");
Serial.print(calGyrY,4); Serial.print(",");
Serial.print(calGyrZ,4); Serial.print(",");
Serial.print(calAccX,2); Serial.print(",");
Serial.print(calAccY,2); Serial.print(",");
Serial.print(calAccZ,2);
Serial.println(">");
}

```

ii. Flex Sensors (2.2 inch)

Designed for use in flex sensing applications, the 2.2 inch flex sensor is a passive resistive measurement sensor whose resistance will rise with flexing, ranging from about 10k Ω when flat to as high as 110k Ω when bent at 90°. All the sensors are connected to a voltage divider consisting of a 5V to GND fixed 10k Ω resistor. The analog voltage at the mid-point is input to one of the analog pins on an Arduino and converted to a 10-bit number (0 - 1023). Five circuits are implemented, one per finger, on pins A0, A1, A2, A3, and A6.



Figure 2 2.2-Inch Resistive Flex Sensor

iii. MPU6050 Inertial Measurement Unit

The MPU6050 is a single chip 3 axis MEMS accelerometer and 3 axis MEMS gyroscope. It communicates through the I2C protocol, at address 0x68 (AD0 is pulled to GND).

Accelerometer records linear acceleration (g-force) in X, Y and Z axes. The gyroscope gives out angular velocity in degrees per second. These six values together represent full 3D hand orientation and motion, which is necessary for ASL signs that are only different in their orientation. The module can be powered from 5V through the on-board LDO regulator (VCC).



Figure 3 MPU6050 6-axis IMU module

iv. HC-05 Bluetooth Module

The HC-05 is a Bluetooth 2.0 SPP module that establishes a transparent wireless UART bridge between the Arduino Nano and a paired Android smartphone. It communicates via UART at 9600 baud using SoftwareSerial on D10 (RX) and D11 (TX). Since HC-05 RXD logic is 3.3V, a $1\text{k}\Omega + 2\text{k}\Omega$ voltage divider on D11 protects it from 5V logic overvoltage. On the Android side, the Silent Voice application reads packets from the paired Bluetooth socket.



Figure 4 HC-05 Bluetooth 2.0 SPP

v. Power Supply System

The glove uses a rechargeable 3.7V 18650 Li-ion battery as its portable power source, allowing untethered operation without a USB cable. The battery connects to a TP4056 USB-C charging and protection module, which handles safe charging at 1A and includes over-charge, over-discharge, and short-circuit protection. The battery output then feeds a DC-DC boost converter module that steps up the 3.7V battery voltage to a stable regulated 5V output. This 5V rail powers the Arduino Nano (via VIN pin), the HC-05 module, the MPU6050, and the flex sensor voltage

divider circuits. Figure 5 shows the three power system components.



(a) TP4056 USB-C Charger



(b) 3.7V Li-ion Battery



(c) DC-DC Boost Converter

Figure 5 Power Supply System / PCB

vi. Perfboard

All circuit elements are mounted and soldered onto a single side copper perfboard as permanent mounting base. The perfboard is used in place of the prototype breadboard in the final assembly for reliable solder joints and compactness to fit into the glove design. The layout of the components has been designed to keep wiring lengths short and to prevent analog flex sensor wiring from interfering with digital I2C or UART signal wiring.



Figure 6 Perfboard used for permanent

vii. Base Glove

All five flex sensors are attached to a plain cotton knit glove that is the physical support for the sensors. This glove offers a comfortable, conforming surface that has space for the natural motion of the hand and maintains the alignment of sensors with the finger joints. Flex sensors are sewn/taped down the back of each finger and wires are connected to the perfboard attached to the back of the hand.



Figure 7 Cotton knit base glove

Table 1 provides a complete summary of all hardware components, their specifications, quantities, and roles.

Table 1 ASL Sensor Glove — Complete Hardware Component List

#	Component	Specification	Qty	Role in System	Interface
1	Arduino Nano	ATmega328P, CH340, 5V, 16MHz, 8 analog pins	1	Main MCU — reads all sensors, builds and transmits data packets	Digital + Analog
2	Flex Sensor (2.2 inch)	~10k Ω flat, ~110k Ω fully bent	5	One per finger — detects individual finger bend angle	Analog ADC
3	MPU6050 IMU	3-axis accel + gyro, I2C addr 0x68, 3.3–5V	1	Captures hand orientation and 3D motion	I2C (A4/A5)
4	HC-05 Bluetooth	BT 2.0 SPP, UART, 5V, 9600 baud	1	Wireless serial bridge between Nano and Android smartphone	UART (D10/D11)
5	Li-ion Battery (3.7V)	3.7V, 2600mAh typical, cylindrical 18650	1	Portable power supply for untethered glove operation	Power
6	TP4056 Charger Module	USB-C input, 5V, 1A charge current, Li-ion	1	Battery charging and protection circuit	Power
7	Boost Converter Module	USB-C input, 5V output regulated, DC-DC boost	1	Boosts 3.7V battery to stable 5V for Arduino and modules	Power
8	Perfboard / PCB	Single-sided, copper pads, 5×7cm	1	Permanent component mounting and soldering base	—
9	Resistor 10k Ω	1/4W, $\pm 5\%$ tolerance	5	Voltage divider — one per flex sensor	Passive
10	Cotton Glove	Plain knit, one size fits most	1	Base glove on which sensors are mounted	—
11	Jumper Wires	Male-to-male and male-to-female	~30	All interconnections between components	—

Table 2 provides the complete pin-level wiring reference for all connections to the Arduino Nano.

Table 2 Complete Arduino Nano Pin Mapping

Component	Component Pin	Nano Pin	Wire Color	Notes
Flex 1 (Thumb)	Divider mid-point	A0	Yellow	Voltage divider output to 10-bit ADC

Flex 2 (Index)	Divider mid-point	A1	Orange	Same circuit
Flex 3 (Middle)	Divider mid-point	A2	Green	Same circuit
Flex 4 (Ring)	Divider mid-point	A3	Blue	Same circuit
Flex 5 (Pinky)	Divider mid-point	A6	Purple	A4/A5 reserved for I2C — must not be used
MPU6050	VCC	5V	Red	Onboard regulator steps to 3.3V internally
MPU6050	GND	GND	Black	Common ground rail
MPU6050	SDA	A4	White	I2C data — dedicated, do not share
MPU6050	SCL	A5	Gray	I2C clock — dedicated, do not share
MPU6050	AD0	GND	Black	Sets I2C address to 0x68
HC-05	VCC	5V	Red	5V from boost converter output
HC-05	GND	GND	Black	Common ground rail
HC-05	TXD	D10 (RX)	Green	HC-05 transmits → Nano SoftwareSerial RX
HC-05	RXD	D11 (TX)	Blue	Use 1kfi+2kfi divider for 3.3V logic safety
Boost Converter	VOUT	5V (to VIN)	Red	Regulated 5V to Nano VIN pin and HC-05
Battery	+3.7V	IN	Red	Li-ion cell → TP4056 → Boost converter input

Figure 8 shows the complete wiring diagram including all sensor connections, I2C bus, SoftwareSerial UART, and power rails.

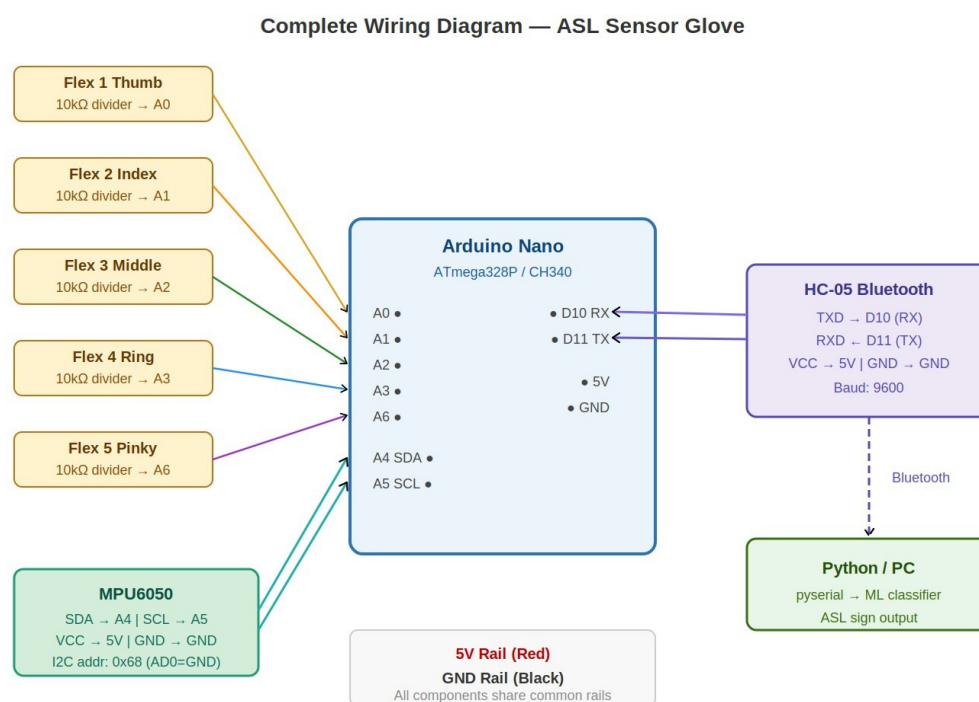


Figure 8 Complete wiring diagram

3.2.3 Software/Tools

The following software tools and frameworks were used in the development of the Silent Voice application and the machine learning model:

Table 3 Software Tools and Frameworks Used

Tool / Framework	Purpose
Android Studio (Kotlin)	IDE for Android app development
Kotlin	Primary programming language for the app
TensorFlow Lite 2.14.0	On-device ML inference engine
Google Colab (Python)	Model training environment
TensorFlow / Keras	Building and training the LSTM model
scikit-learn	Label encoding, train/test split, StandardScaler
NumPy / Pandas	Data loading, preprocessing, and downsampling
Room Database (SQLite)	Local gesture history storage on Android
Android TextToSpeech API	Speech output for recognized gestures
Bluetooth Classic (RFCOMM)	Wireless communication between glove and app
ASL Sensor Dataglove Dataset	Public CSV dataset used for model training

3.3 Algorithms, Model Training and Implementation Details

3.3.1 Dataset and Feature Selection

The training data set used was ASL Sensor Dataglove Dataset which is publicly available. A total of 30 classes were selected: 26 letters (A–Z) and 4 common ASL words (yes, no, me, you). The data samples each include 11 sensor features: 5 flex sensors and 3 axes of gyroscope sensors and 3 axes of accelerometer sensors.

3.3.2 Data Preprocessing

Raw CSV sequences (up to 1500 time steps) were downsampled to 100 frames using linear interpolation via `numpy.linspace`. Data was then normalised using `StandardScaler` and split 80/20 into training and test sets using `sklearn's train_test_split` with random seed 42.

3.3.3 LSTM Model Architecture

A two-layer LSTM network was designed to capture the temporal dynamics of glove sensor readings. The final model architecture is shown in Table 4 below.

Table 4 LSTM Model Architecture

Layer	Configuration	Output Shape
Input	shape = (100, 11)	(None, 100, 11)
LSTM (1st)	128 units, return_sequences=True	(None, 100, 128)
Dropout	rate = 0.3	(None, 100, 128)
LSTM (2nd)	64 units	(None, 64)
Dropout	rate = 0.3	(None, 64)
Dense (Hidden)	64 units, ReLU activation	(None, 64)
Dense (Output)	30 units, Softmax activation	(None, 30)

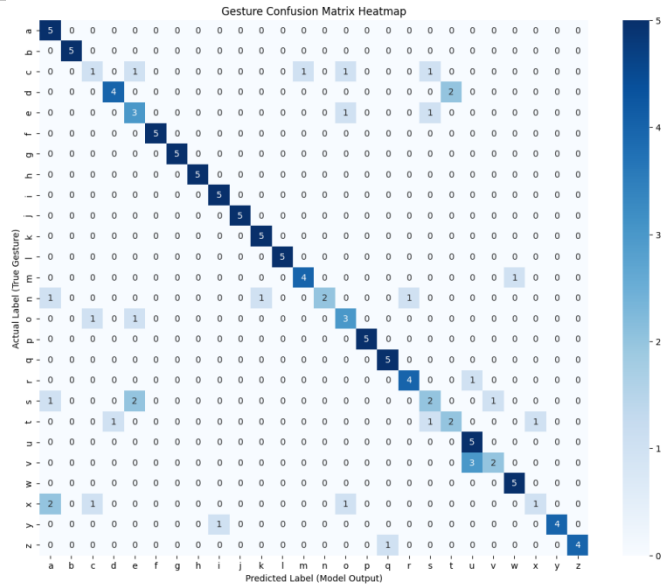


Figure 9 Model Matrix

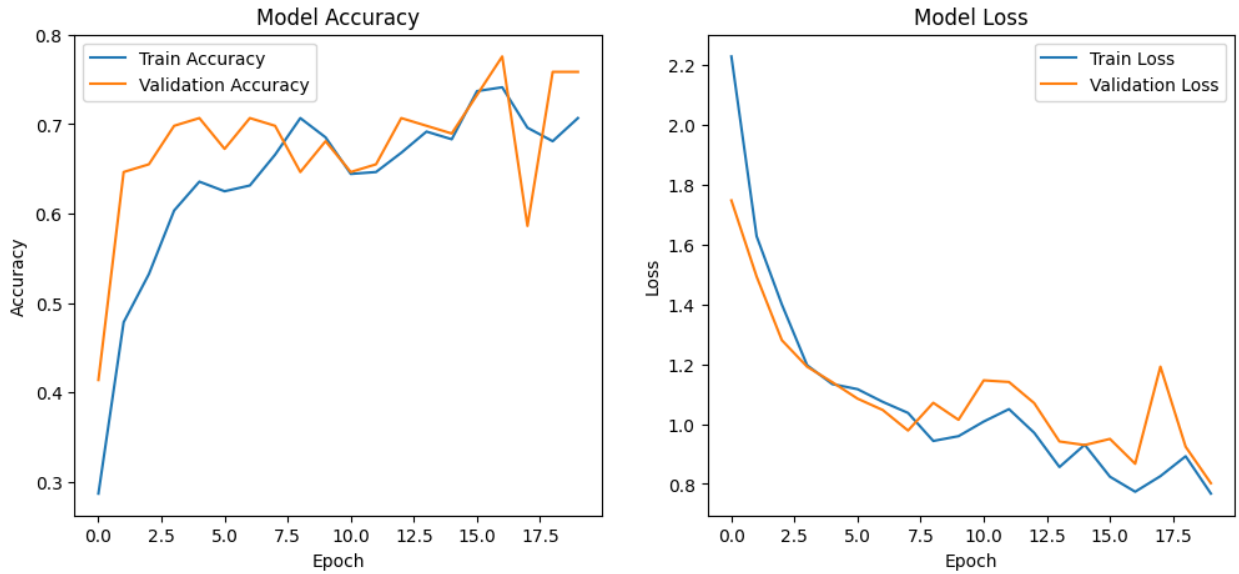


Figure 10 Model accuracy & loss gesture A

Table 5 PER-CLASS F1-SCORES

	Precision	Recall	f1-score	Support
a	0.56	1.00	0.71	5
b	1.00	1.00	1.00	5
c	0.33	0.20	0.25	5
d	0.80	0.67	0.73	5
e	0.43	0.60	0.50	5
f	1.00	1.00	1.00	5
g	1.00	1.00	1.00	5
h	1.00	1.00	1.00	5
i	0.83	1.00	0.91	5
j	1.00	1.00	1.00	5
k	0.83	1.00	0.91	5
l	1.00	1.00	1.00	5
m	0.80	0.80	0.80	5
n	1.00	0.40	0.57	5
o	0.50	0.60	0.55	5
p	1.00	1.00	1.00	5
q	0.83	1.00	0.91	5
r	0.80	0.80	0.80	5
s	0.40	0.33	0.36	5
t	0.50	0.40	0.44	5
u	0.56	1.00	0.71	5
v	0.67	0.40	0.50	5
w	0.83	1.00	0.91	5
x	0.50	0.20	0.29	5
y	1.00	0.80	0.89	5
z	1.00	0.80	0.89	5
accuracy			0.77	132
Macro avg	0.78	0.77	0.76	132
Weighted avg	0.77	0.77	0.75	132

3.3.4 Bluetooth Communication and Data Protocol

A Bluetooth Classic communication was realized by using the BluetoothSocket API of Android with RFCOMM serial profile and in Kotlin. When 'Connect Bluetooth' is pressed, the app starts a background thread which opens a socket to the HC-05 module. The app will automatically default to an insecure socket if the secure connection fails. The glove sends frames in the format <v1,v2,...v11>. The app receives incoming bytes, decodes entire packets and outputs 11 float values per frame.

3.3.5 Real-Time Prediction Pipeline

A rolling buffer of 100 frames is maintained of incoming frames. A call to the LSTM model is made every 10 new frames. Only predictions with a confidence greater than 75% are accepted: (1) confidence greater than 75%, and (2) the same gesture is repeated in three or more runs. It is then demonstrated, spoken through TTS and stored on the Room database. A motion detection filter will

empty the buffer at a hand at rest (flex variance < 100).

3.4 Mobile Application

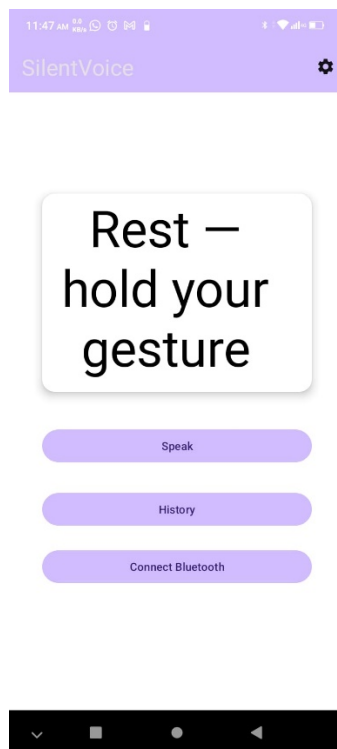


Figure 11 Mobile app front page

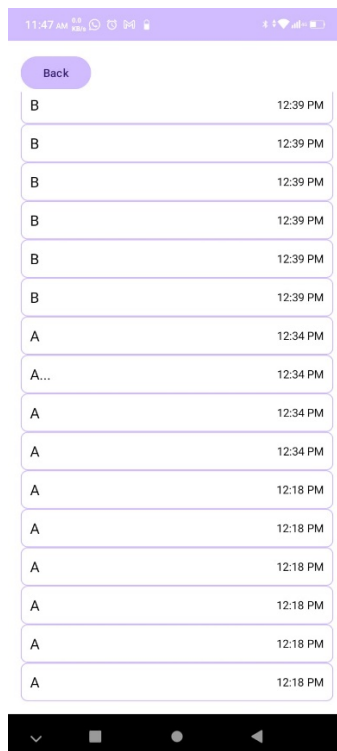


Figure 12 History

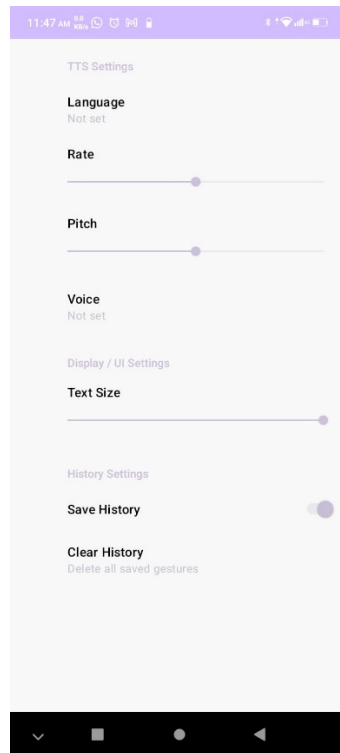


Figure 13 App Settings

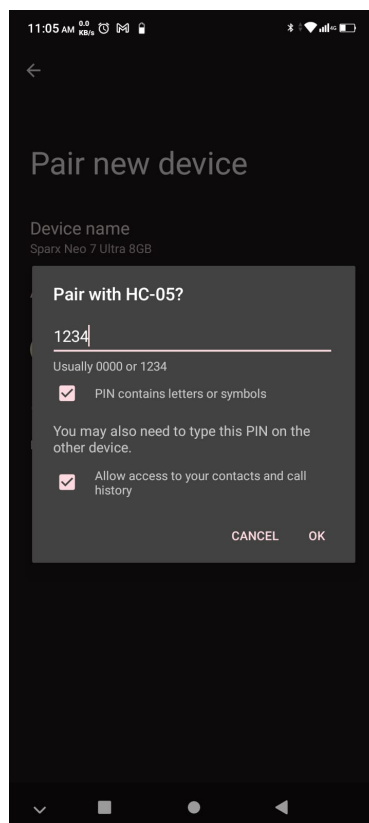


Figure 14 App pair with Bluetooth

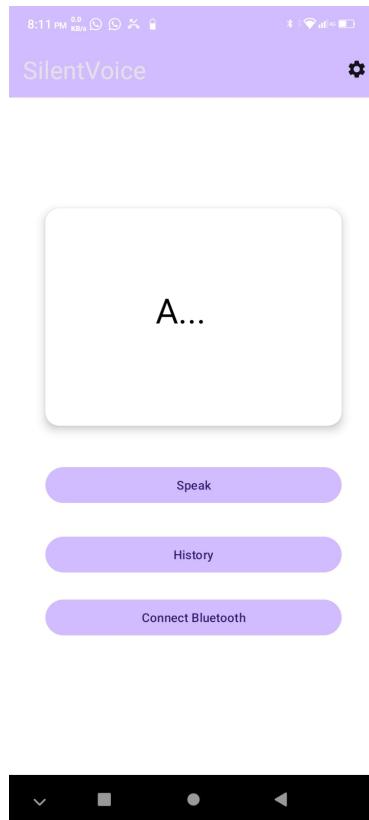


Figure 15 Gesture Dectecting

Chapter 4

Testing and Validation/Discussion

The testing, validation and results of the Silent Voice ASL Sign Language Translation Glove system are presented in this chapter. It includes hardware prototype test results, functional test cases, model accuracy and live gesture prediction results during end-to-end system test. The details of verification, testing and validation processes of the project, statistical analysis, calculations, results and evidences are included in this chapter, covering the following key points:

- The system and hardware design meets the requirements. Verification of the algorithm design and/or data.
- The specification of hardware and/or algorithm tests has been verified.
- The verification of data used. The test cases for each level of design and the integration of design models to the next level of design were tested with the project requirement and project objective.
- Validation & analysis of desired results & outcomes.

4.1 Testing

The Silent Voice system hardware prototype was built using a copper perfboard and attached to a cotton knit glove. Each finger joint had 5, 2.2-inch, resistive flex sensors attached to the top of the finger. MPU6050 IMU was fastened at the back of the hand for capturing the hand 3D orientation and motion. Arduino Nano microcontroller, HC-05 Bluetooth module, TP4056 charger and DC-DC boost converter were soldered to the perfboard and powered by 3.7V 18650 LION Battery. Assembled prototype was then tested for electrical conduction and mechanical strength before functional testing. The completed prototype of the glove for live gesture prediction testing is as shown in Figure 8.

4.1.1 Prototypes

The Silent Voice system hardware prototype was assembled on a copper perfboard and mounted on a cotton knit glove. Five 2.2-inch resistive flex sensors were placed along the dorsal side of each finger joint. The MPU6050 IMU was secured on the back of the hand to capture 3D orientation and motion. The Arduino Nano microcontroller, HC-05 Bluetooth module, TP4056 charger, and DC-DC boost converter were soldered onto the perfboard and powered by a 3.7V 18650 Li-ion battery. The assembled prototype was verified for electrical continuity and physical durability before proceeding to functional testing. Figure 8 shows the completed glove prototype during live gesture prediction testing.

4.1.2 Test Cases

A total of ten functional test cases were defined to verify end-to-end system behavior, covering

hardware communication, data transmission, model inference, and application output. Table 5 presents the complete test case results.

Table 6 Functional Test Cases and Results

No.	Test Case	Input	Expected Output	Actual Output	Result
TC-01	Bluetooth Connection	Power on glove, open app, press Connect Bluetooth	App connects to HC-05 successfully	App connected to HC-05	Pass
TC-02	Sensor Data Transmission	Glove powered and connected	11 float values received per frame in format <v1,...,v11>	Data packets received correctly	Pass
TC-03	Frame Buffering	Continuous sensor data stream	100 frames buffered before inference	Buffer fills to 100 frames	Pass
TC-04	Gesture A Prediction	Perform ASL gesture A (closed fist)	App displays "A" with confidence $\geq 75\%$	App displayed "A" correctly	Pass
TC-05	Gesture B Prediction	Perform ASL gesture B (four fingers up)	App displays "B" with confidence $\geq 75\%$	App displayed "B" correctly	Pass
TC-06	Gesture F Prediction	Perform ASL gesture F (okay sign)	App displays "F" with confidence $\geq 75\%$	App displayed "F" correctly	Pass
TC-07	Text-to-Speech Output	Gesture correctly predicted	Predicted gesture spoken aloud	TTS engine spoke predicted letter	Pass
TC-08	False Positive Prevention	Hand at rest / no gesture	No prediction displayed	No output shown during rest	Pass
TC-09	Gesture History Logging	Gesture predicted successfully	Gesture saved in Room database with timestamp	History entry saved and visible	Pass
TC-10	Offline Operation	No internet connection	System operates fully offline	All features worked without internet	Pass

4.2 Results and Output

This section shows the results achieved by end-to-end testing of the Silent Voice system, including the accuracy of model training, real-time gesture prediction in the application and correctness of the application output.

4.2.1 Completion

All six project objectives outlined in Chapter 1 were accomplished. The setup and calibration of the wearable sensor glove was done. The glove and the Android application were able to communicate reliably using a Bluetooth RFCOMM communication link. The LSTM model was trained using the ASL Sensor Dataglove Dataset and then was converted to a TensorFlow Lite model to be deployed on the Android. The Silent Voice Android application was built out with live presentation of gestures, Text-to-Speech output, gesture history logging, and a settings screen which can be customized. The 10 functional test cases were successfully completed in end-to-end testing, which proves the completion of the system.

4.2.2 Accuracy

The LSTM model was trained for 50 epochs on the ASL Sensor Dataglove Dataset using 80% of the data for training and 20% for testing. The model was finally tested and scored an accuracy of 77.93% and macro-average F1-score of 0.76 for all the 30 gesture classes. The gestures A, B and F were correctly predicted with high level of confidence, as verified during the live end to end testing of the complete system, on the Android application. The live gesture prediction results obtained during the testing of the system are shown in Table 6. The results of the live gesture prediction are shown in Table 7.

Table 7 Live Gesture Prediction Results

Gesture	ASL Description	Expected Prediction	Actual Prediction	Confidence	Result
<i>A</i>	Closed fist, thumb resting on side	<i>A</i>	<i>A</i>	$\geq 75\%$	Correct
<i>B</i>	Four fingers straight up, thumb tucked across palm	<i>B</i>	<i>B</i>	$\geq 75\%$	Correct
<i>F</i>	Index finger and thumb form circle, other fingers up	<i>F</i>	<i>F</i>	$\geq 75\%$	Correct

Figures 11, 12, and 13 show screenshots of the Silent Voice Android application displaying correctly predicted gestures A, B, and F respectively during live testing.

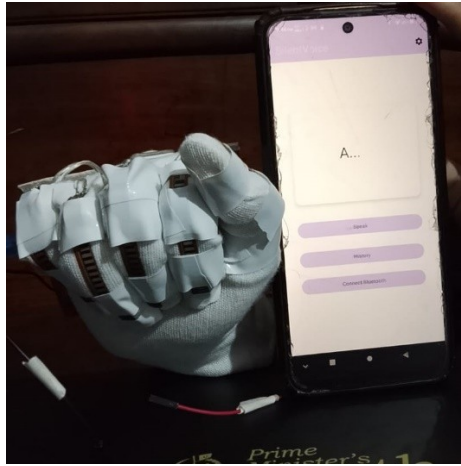


Figure 16 Silent Voice App — Gesture A Prediction

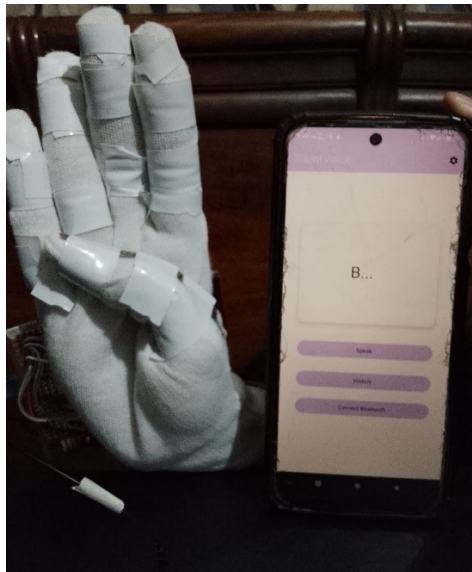


Figure 17 Silent Voice App — Gesture B Prediction

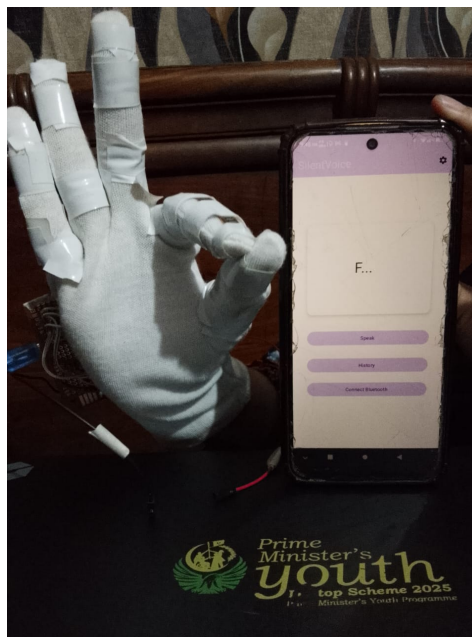


Figure 18 Silent Voice App — Gesture F Prediction

4.2.3 Correctness

The validity of the system was checked by doing an end-to-end testing of the system, that is, when the physical ASL gestures were made on the glove, it was properly recognized and displayed on the Android application. During live sessions, this system was correct in prediction of gestures A, B and F with the prediction being outputted in the form of text and audio through the Text to Speech engine. The testing of the confidence threshold (75%) and three confirmation filter were effective in avoiding false positives. Everything from Bluetooth communication to parsing sensor data, making LSTM inferences and producing TTS sounds was working fine according to the system requirements described in Chapter 3.

Chapter 5

Conclusion and Future Recommendations

This chapter summarizes the work completed, reflects on the objectives achieved and provides recommendations for future development and extension of the capability and impact of the system in the Silent Voice Sign Language Translation Glove project.

5.1 Conclusion

The project was successful in designing and implementing a complete end to end wearable assistive communication system for American Sign Language Recognition. The system has two main parts: the sensor glove with 5 resistive flex sensors and one IMU (MPU6050), which is controlled by an Arduino Nano microcontroller that sends the real-time data from the hand gestures to the Android application through wireless communication (HC-05 Bluetooth Classic module), while the Android application, developed in Kotlin, receives the data from the sensors, processes it with a LSTM model compiled in TensorFlow Lite running on the device and displays the recognized gestures as text and English speech. The six project objectives have been achieved. It is successfully fabricated and calibrated to be worn on the hand. A stable Bluetooth RFCOMM communication was achieved between the glove and the Android device. The ASL Sensor Dataglove Dataset was used to train a two-layer LSTM to classify 30 gesture classes (A–Z plus yes, no, me, you), the macro-average F1 score for each class was 0.76, and the final test accuracy was 77.93%. Model converted to TensorFlow Lite and deployed on Android device for fully offline on-device inference. The Silent Voice application was finished with all the intended functions: real-time gesture display, Text-to-Speech output, and logging of all gestures, including times, and a screen to set or change the settings. All the ten defined test cases were successful with the end-to-end system testing that confirmed the successful operation of all components. The project also shows that it is possible to construct a practical low cost portable ASL translation system without the need to use a camera, any form of server processing or internet connection using only readily available components including the Arduino Nano, HC-05 Bluetooth module and a mid-range Android smartphone. This renders the Silent Voice system accessible and usable in a variety of real-world contexts where the deaf and hard-of-hearing community need to communicate with the majority hearing community.

5.2 Future Recommendations

The Silent Voice system successfully shows the fundamental idea of real time wearable ASL recognition but there are certain directions for future improvement:

- **Extended Vocabulary:** There are currently 30 classes in the system (26 ASL alphabet letters A–Z and 4 common words: yes, no, me, you). Future research would involve expanding the

vocabulary to include common ASL words and phrases outside of the alphabet, allowing for more natural communication in ASL.

- Improved Model Accuracy: Some symbols had lower F1-scores such as T, V, O and S, which are similar in terms of visual features. Data augmentation, class weighting loss functions and attention mechanisms may be considered to boost class performance on these gestures that are confused.
- Larger and More Diverse Dataset: only one publicly available dataset was used to train the model. Data from many individuals with different hand size and signing style should be gathered in order to get a better generalisation of the model and to minimise user dependent variability.
- Bluetooth Low Energy (BLE): The existing one is using Bluetooth Classic (HC-05) which consumes more power. Switching to an off-the-shelf BLE module (HC-10 or nRF52840) would greatly improve the battery life of the glove, which would make it more convenient for all day wear.

Over the Air (OTA) Firmware Updates:

This would be possible if the components were reduced in number, size and power consumption by replacing the Arduino Nano and separate HC-05 module with an ESP32 microcontroller that integrates the processing and Bluetooth/Wi-Fi. Isolation of gesturers: The existing system is based on the isolated gesture recognition. In the future, a sequence-to-sequence translation model may be used to recognize to complete sentences in ASL and to generate grammatically correct sentences in English. Two-Way Communication: It is possible to add a display or haptic feedback module to the glove to enable hearing users to communicate back to the sign language user, creating a truly two way assistive communication.

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